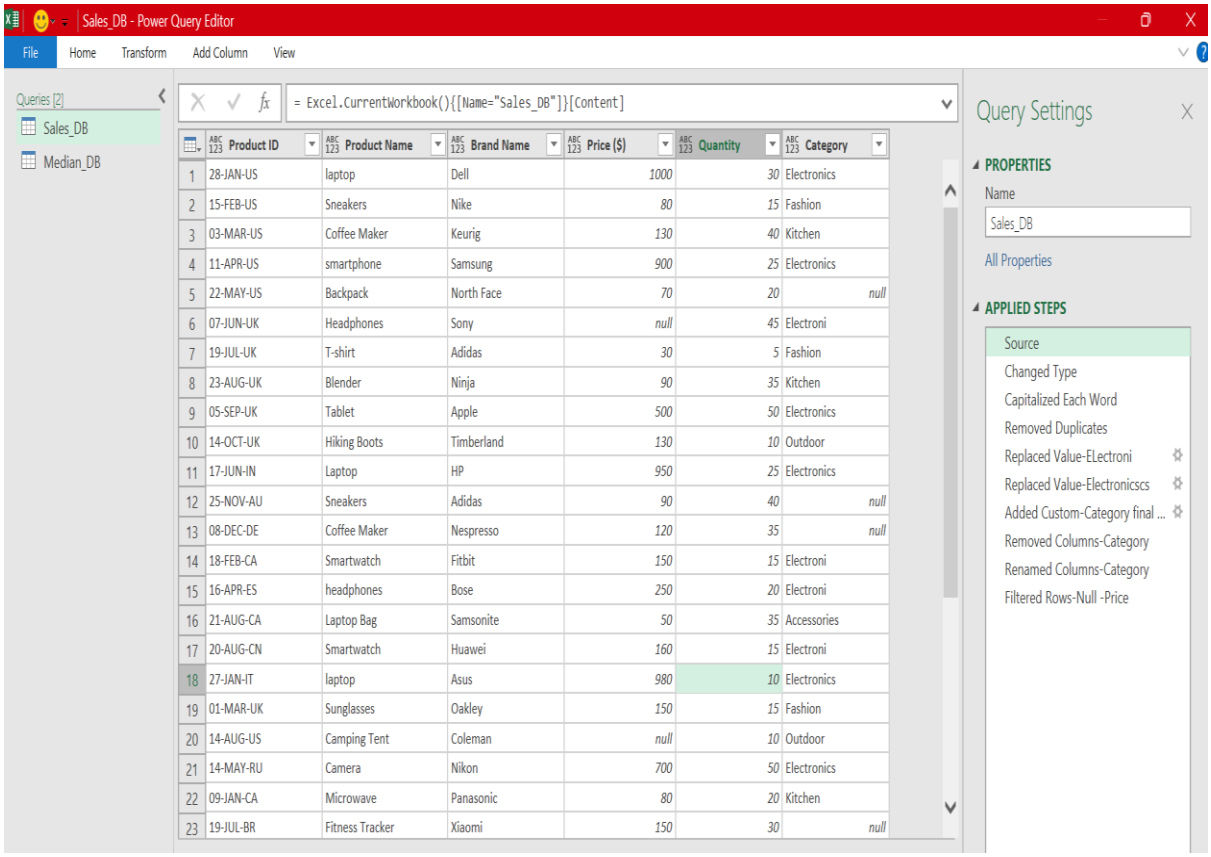
Excel Assignment -2
Data Cleaning and Transformation

Data Cleaning :

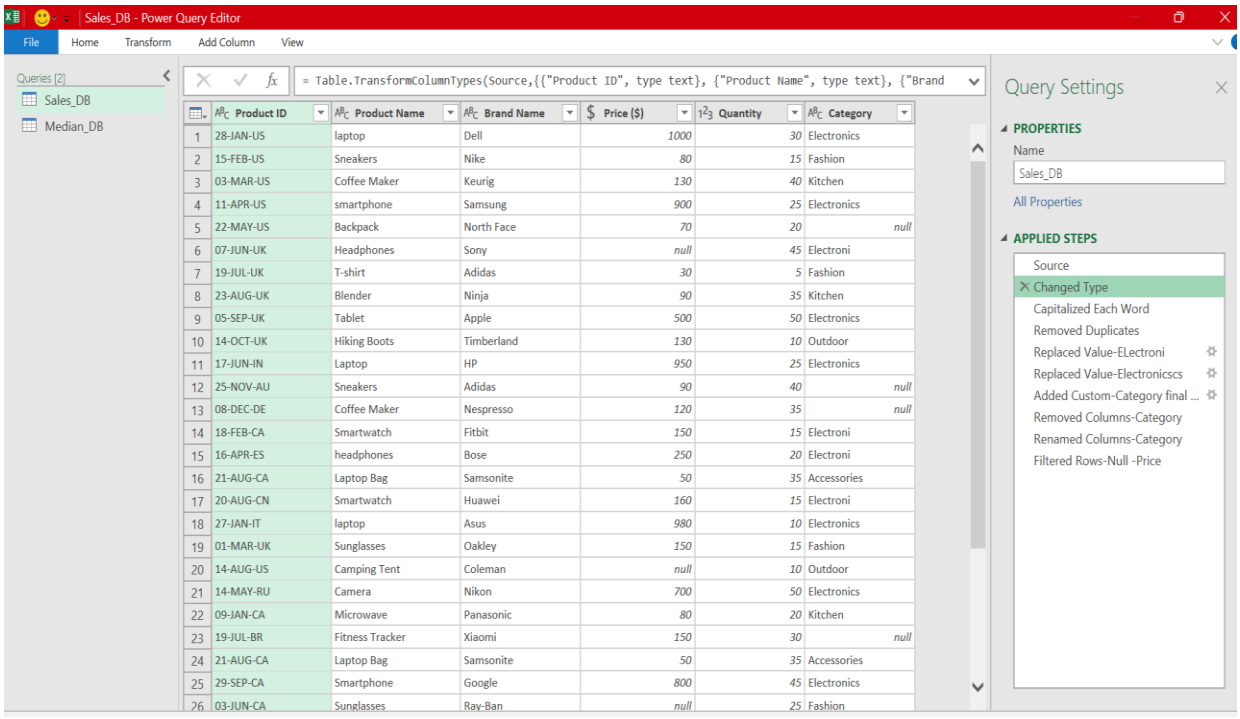
The given Dataset is taken to PowerQuery for further Cleaning and Transformation Process.

1.Data's taken to PowerQuery .



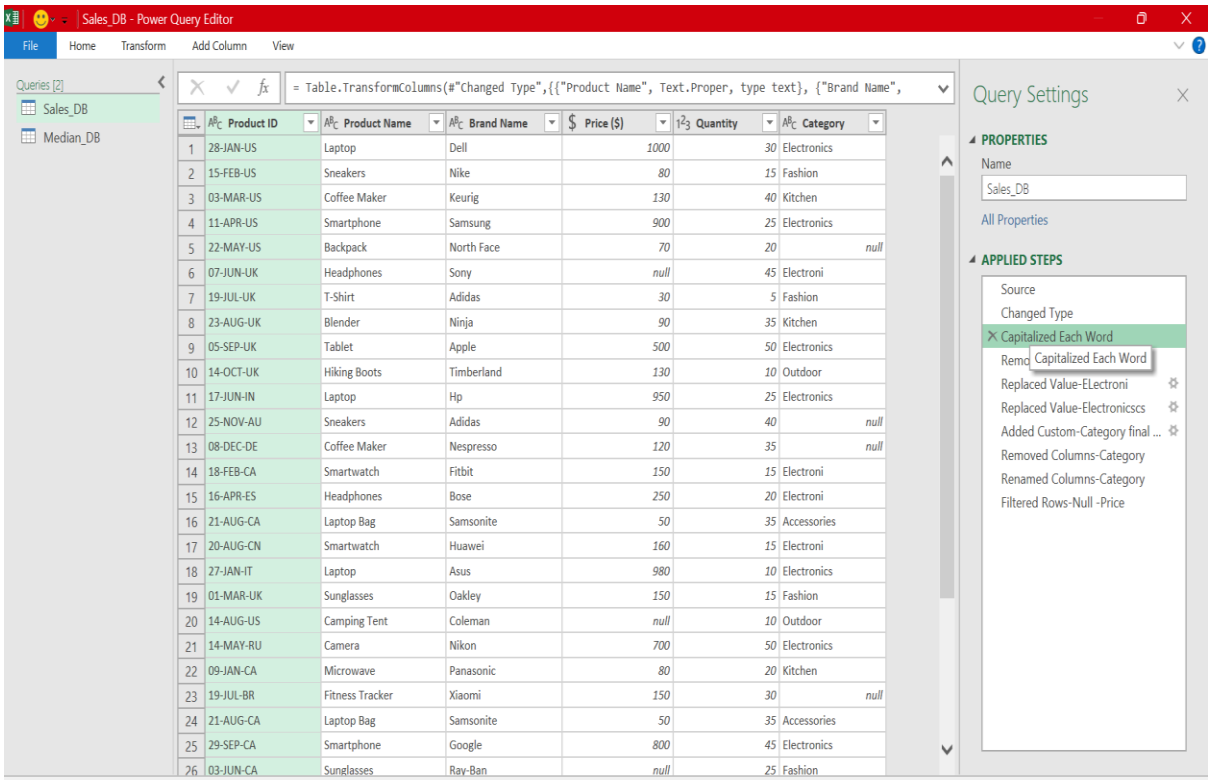
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electroni
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
15	16-APR-ES	headphones	Bose	250	20	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
18	27-JAN-IT	laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null

2.Changed the Column Types accordingly.



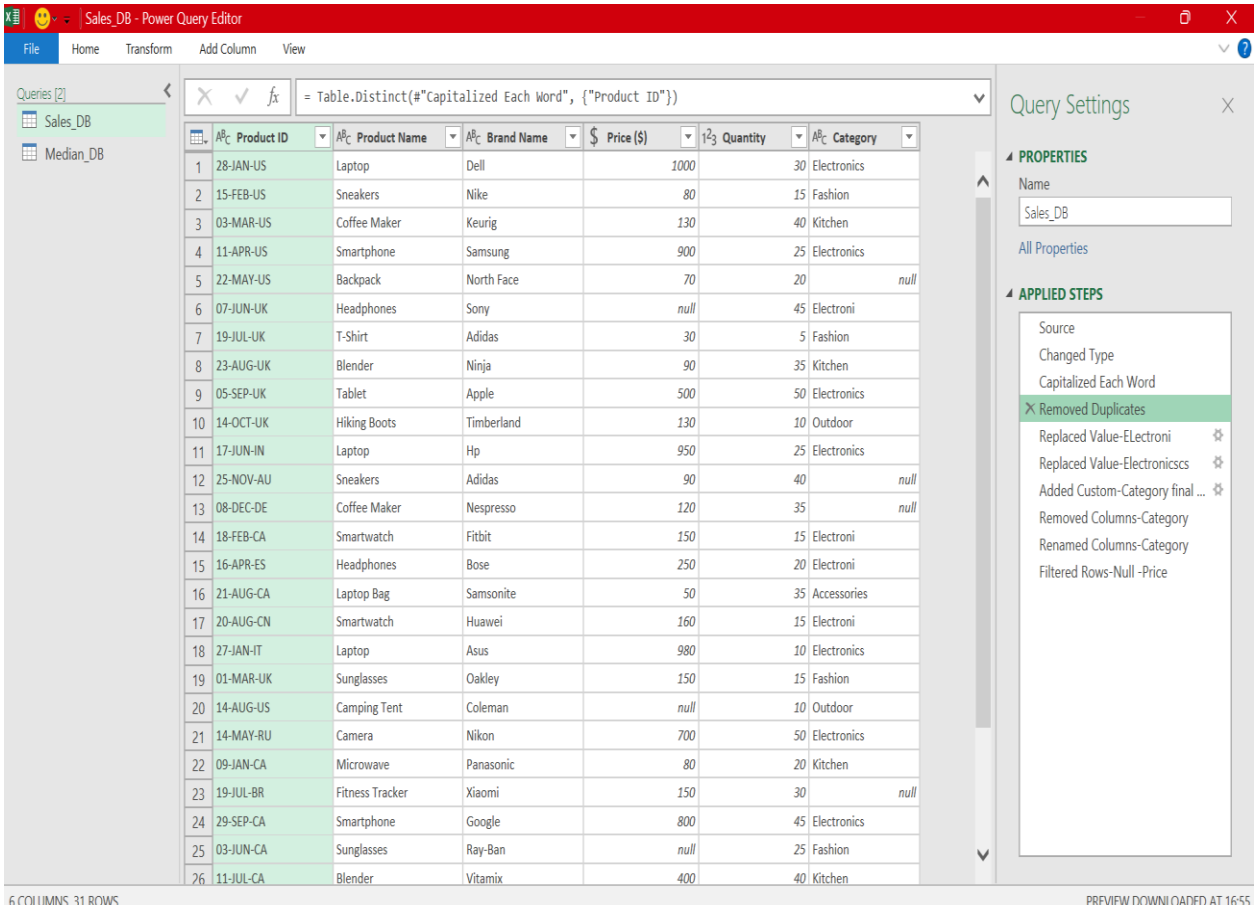
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electroni
7	19-JUL-UK	T-shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	HP	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
15	16-APR-ES	headphones	Bose	250	20	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
18	27-JAN-IT	laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion

3.Arranged the letter formats



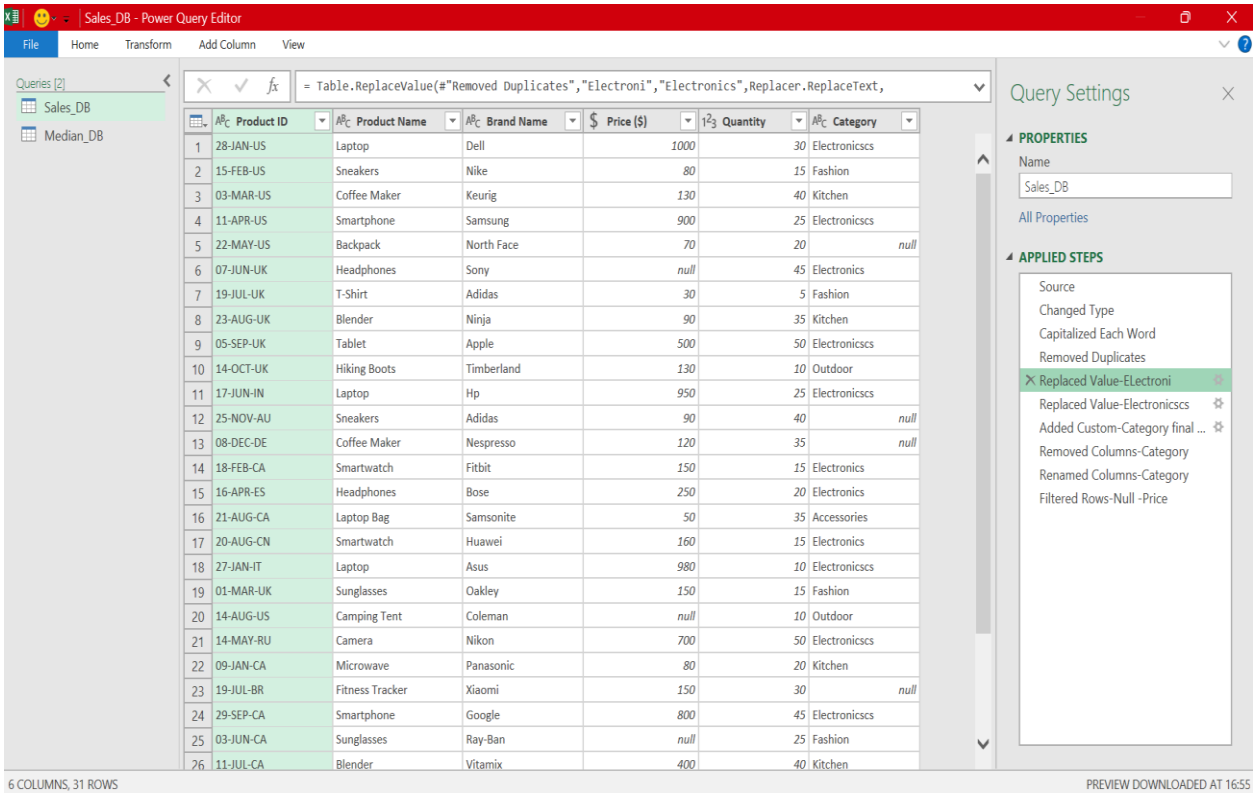
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electroni
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	Hp	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
15	16-APR-ES	Headphones	Bose	250	20	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
18	27-JAN-IT	Laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null
24	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
25	29-SEP-CA	Smartphone	Google	800	45	Electronics
26	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion

4.Removed Duplicates using Product ID



	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electroni
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	Hp	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni
15	16-APR-ES	Headphones	Bose	250	20	Electroni
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni
18	27-JAN-IT	Laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null
24	29-SEP-CA	Smartphone	Google	800	45	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion
26	11-JUL-CA	Blender	Vitamix	400	40	Kitchen

5.Replaced incorrect Spellings.



	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronicscs
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronicscs
5	22-MAY-US	Backpack	North Face	70	20	null
6	07-JUN-UK	Headphones	Sony	null	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronicscs
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	Hp	950	25	Electronicscs
12	25-NOV-AU	Sneakers	Adidas	90	40	null
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronicscs
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronicscs
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null
24	29-SEP-CA	Smartphone	Google	800	45	Electronicscs
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion
26	11-JUL-CA	Blender	Vitamix	400	40	Kitchen

Data Transformation: 1.Handling Missing Values:

1.If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.

- Added Custom coloumn
- Used if [Category] = null then

if [Product Name] = "Backpack" then "Outdoor"

else if [Product Name] = "Sneakers" then "Fashion"

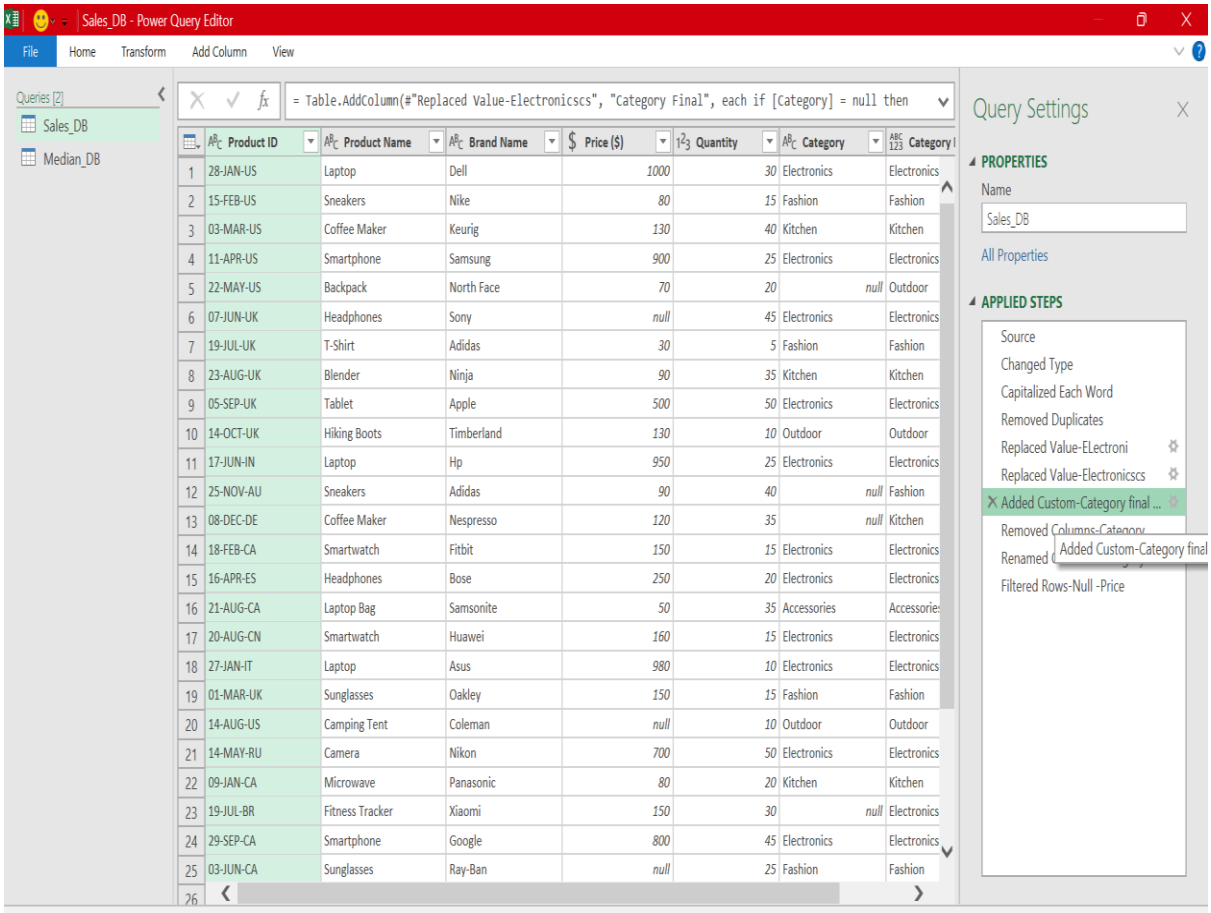
else if [Product Name] = "Coffee Maker" then "Kitchen"

else if [Product Name] = "Fitness Tracker" then "Electronics"

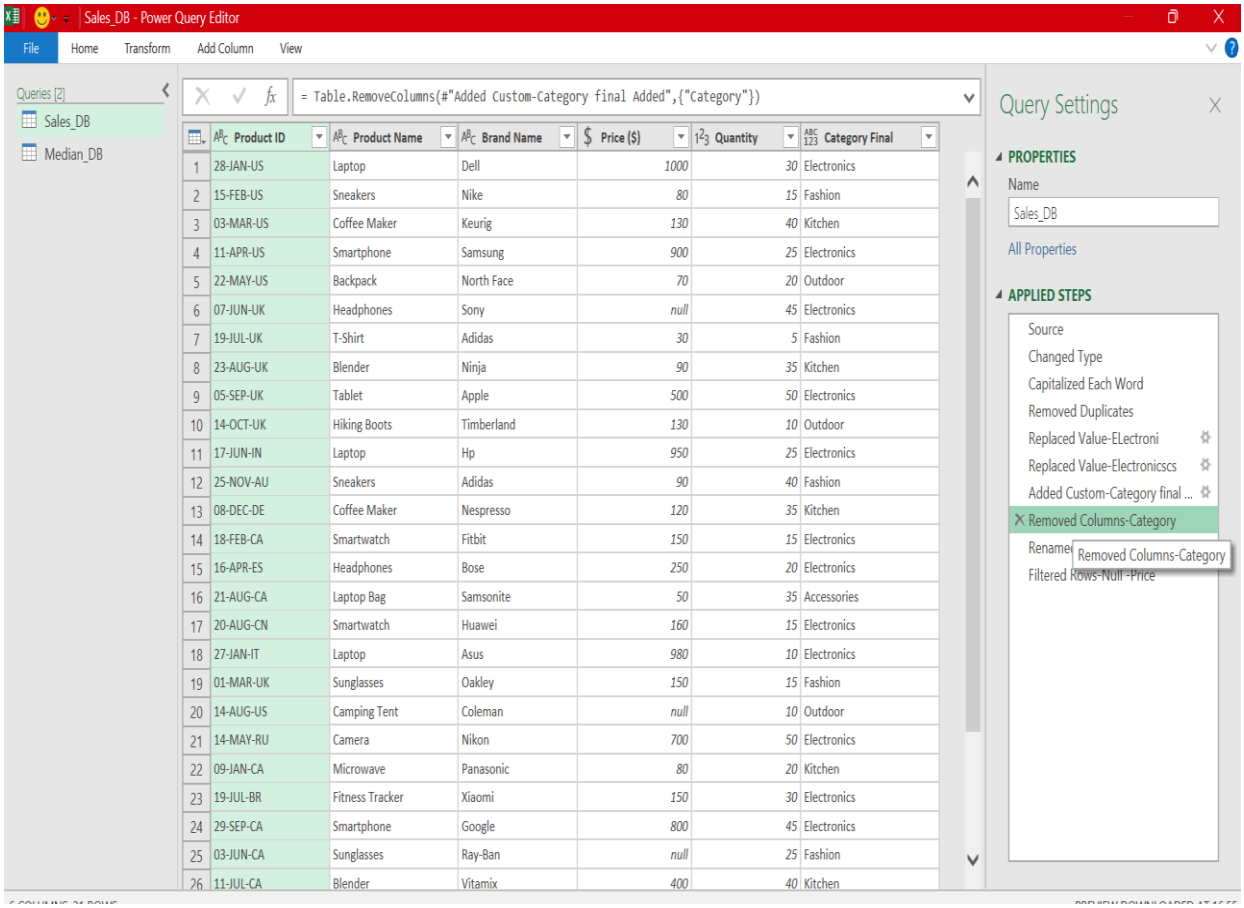
else null

else [Category]

- Removed the Category coloum with null values.



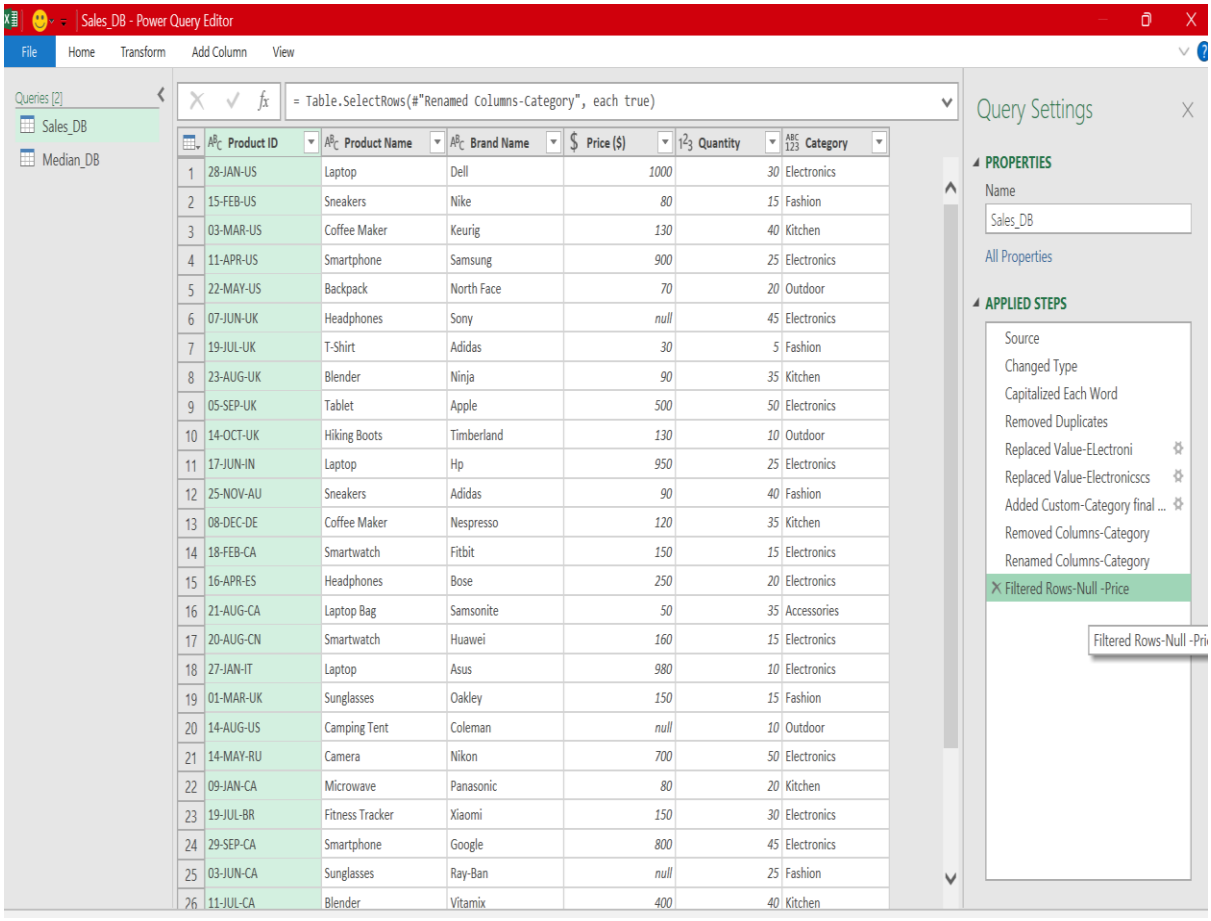
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics	Electronics
5	22-MAY-US	Backpack	North Face	70	20	null	Outdoor
6	07-JUN-UK	Headphones	Sony	null	45	Electronics	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Outdoor
11	17-JUN-IN	Laptop	Hp	950	25	Electronics	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	null	Fashion
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	null	Kitchen
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Electronics
15	16-APR-ES	Headphones	Bose	250	20	Electronics	Electronics
16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	null	Electronics
24	29-SEP-CA	Smartphone	Google	800	45	Electronics	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion	Fashion
26							



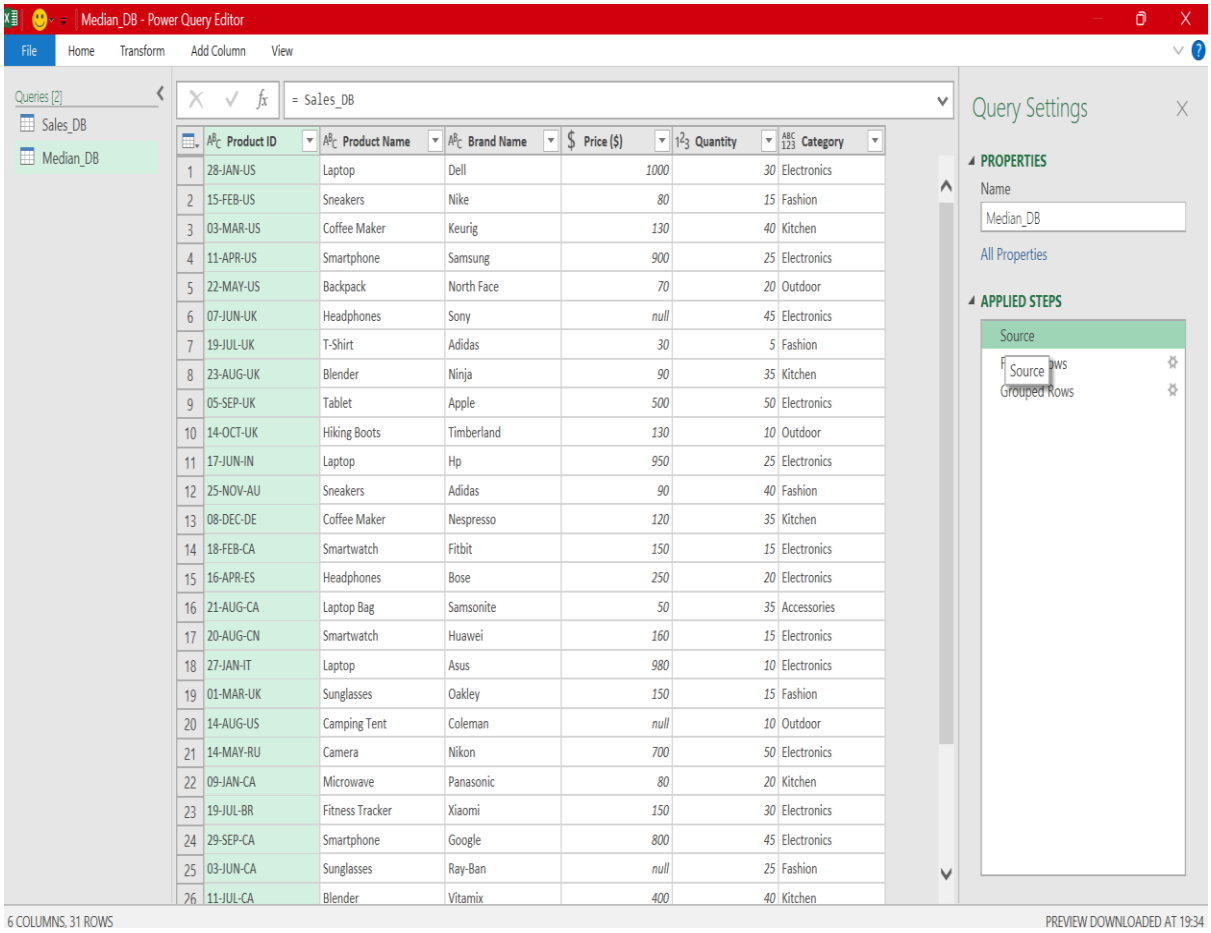
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category Final
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
3	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	Outdoor
6	07-JUN-UK	Headphones	Sony	null	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
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16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
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18	27-JAN-IT	Laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
24	29-SEP-CA	Smartphone	Google	800	45	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion
26	11-JUL-CA	Blender	Vitamix	400	40	Kitchen

2. Check for missing values in the 'Price' column. How would you handle products with missing price information?

- Filtered out Null value in Price

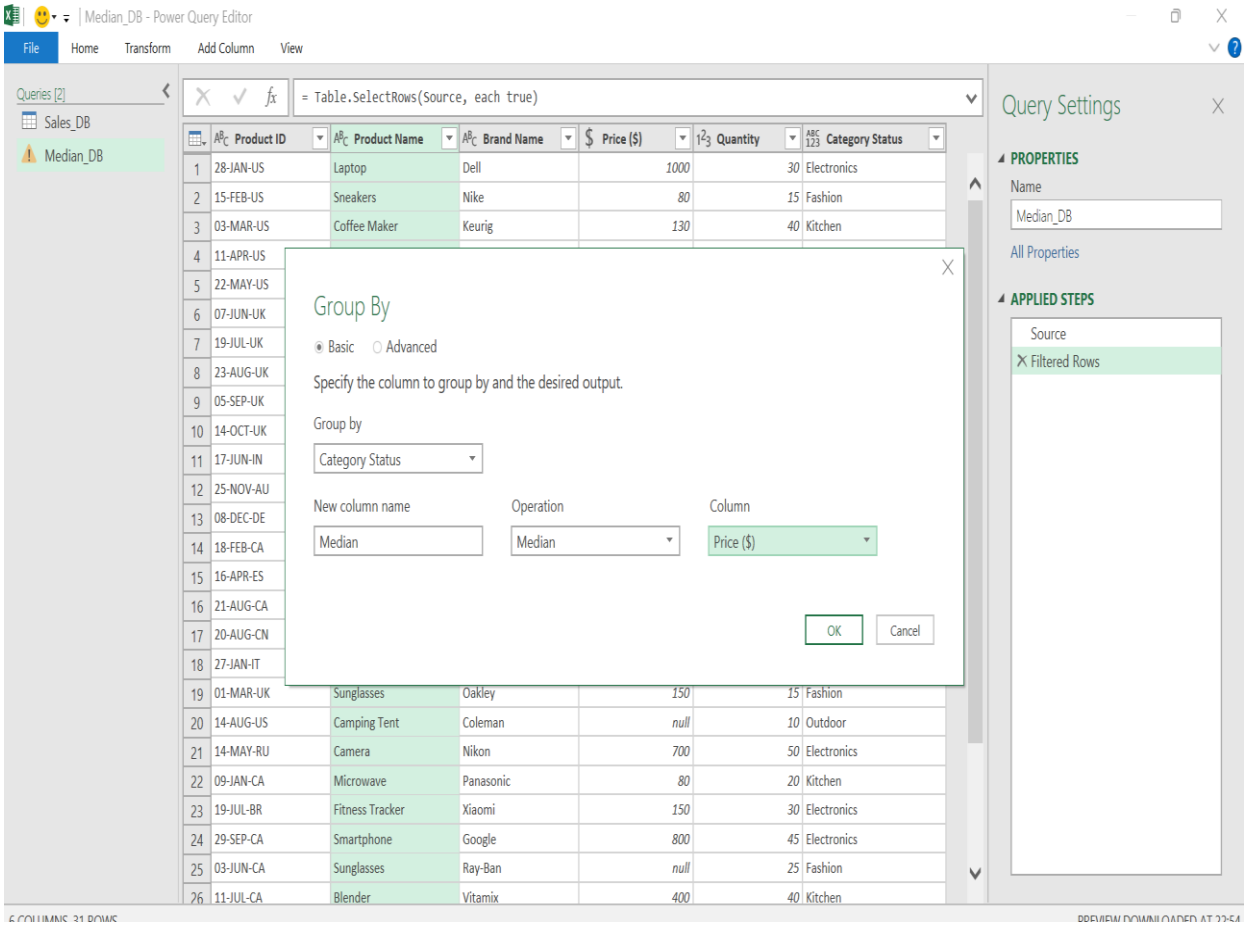


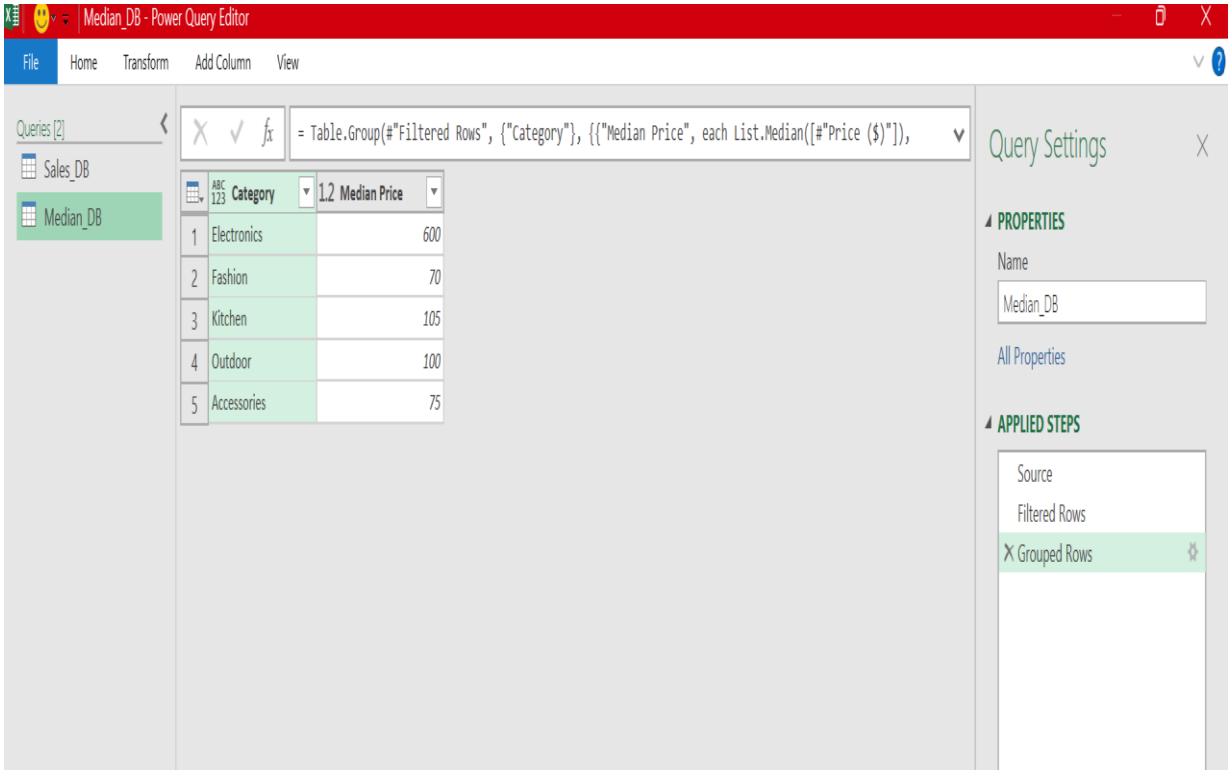
- Created reference PowerQuery named Median_DB



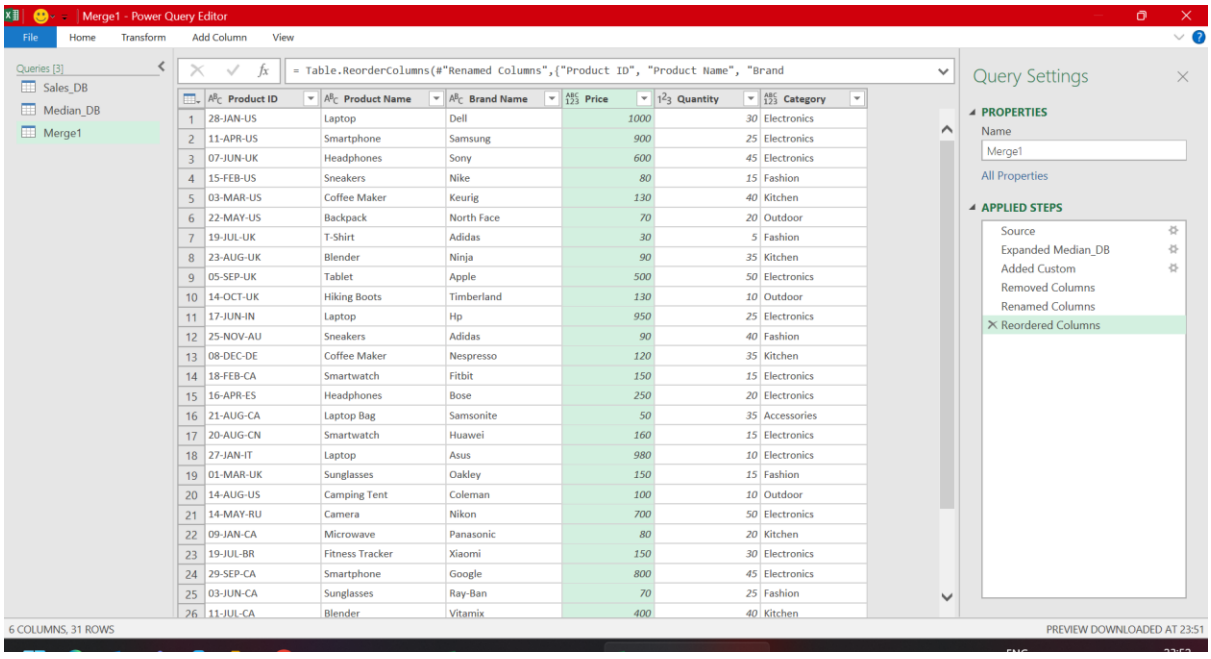
	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
1	28-JAN-US	Laptop	Dell	1000	30	Electronics
2	15-FEB-US	Sneakers	Nike	80	15	Fashion
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4	11-APR-US	Smartphone	Samsung	900	25	Electronics
5	22-MAY-US	Backpack	North Face	70	20	Outdoor
6	07-JUN-UK	Headphones	Sony	null	45	Electronics
7	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion
8	23-AUG-UK	Blender	Ninja	90	35	Kitchen
9	05-SEP-UK	Tablet	Apple	500	50	Electronics
10	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
11	17-JUN-IN	Laptop	Hp	950	25	Electronics
12	25-NOV-AU	Sneakers	Adidas	90	40	Fashion
13	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
14	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
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16	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
17	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
18	27-JAN-IT	Laptop	Asus	980	10	Electronics
19	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
20	14-AUG-US	Camping Tent	Coleman	null	10	Outdoor
21	14-MAY-RU	Camera	Nikon	700	50	Electronics
22	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
23	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
24	29-SEP-CA	Smartphone	Google	800	45	Electronics
25	03-JUN-CA	Sunglasses	Ray-Ban	null	25	Fashion
26	11-JUL-CA	Blender	Vitamix	400	40	Kitchen

- Grouped the category and found the median price.



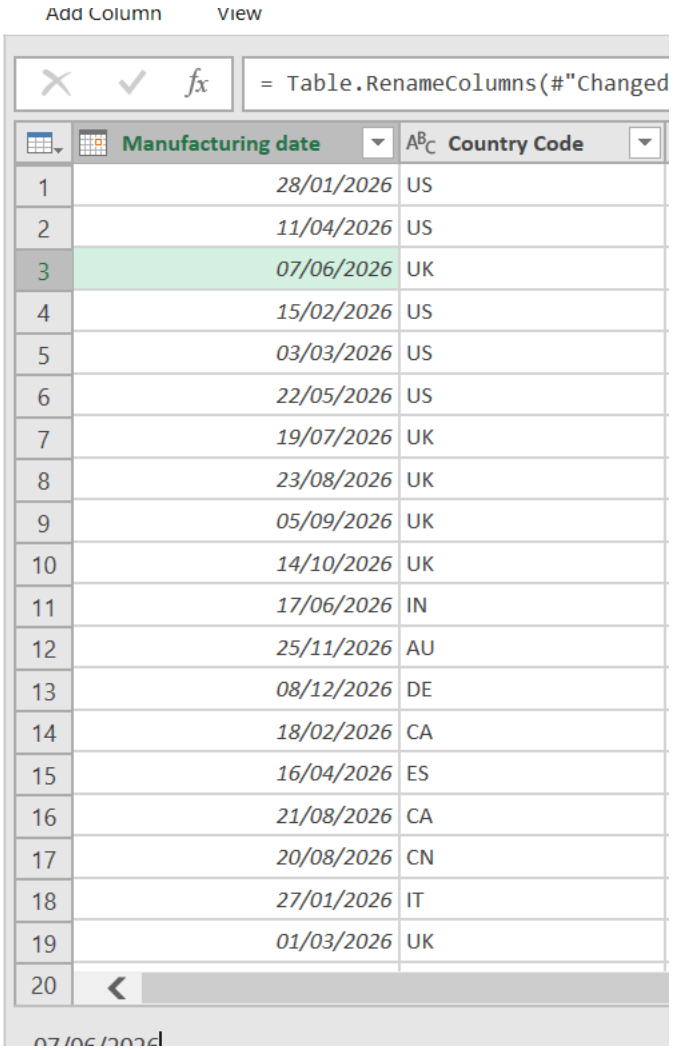


- Merged both powerquery table and replaced missing price values with the median value found.



Splitting and Merging Data:

- Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.



- Merge the "Brand Name" and "Product Name" columns into one column

named "Product Brand".



Product Brand
Laptop-Dell
Smartphone-Samsung
Headphones-Sony
Sneakers-Nike
Coffee Maker-Keurig
Backpack-North Face
T-Shirt-Adidas
Blender-Ninja
Tablet-Apple
Hiking Boots-Timberland
Laptop-Hp
Sneakers-Adidas
Coffee Maker-Nespresso
Smartwatch-Fitbit
Headphones-Bose
Laptop Bag-Samsonite
Smartwatch-Huawei
Laptop-Asus
Sunglasses-Oakley
Camping Tent-Coleman
Camera-Nikon
Microwave-Panasonic
Fitness Tracker-Xiaomi
Smartphone-Google
Sunglasses-Ray-Ban
Blender-Vitamix

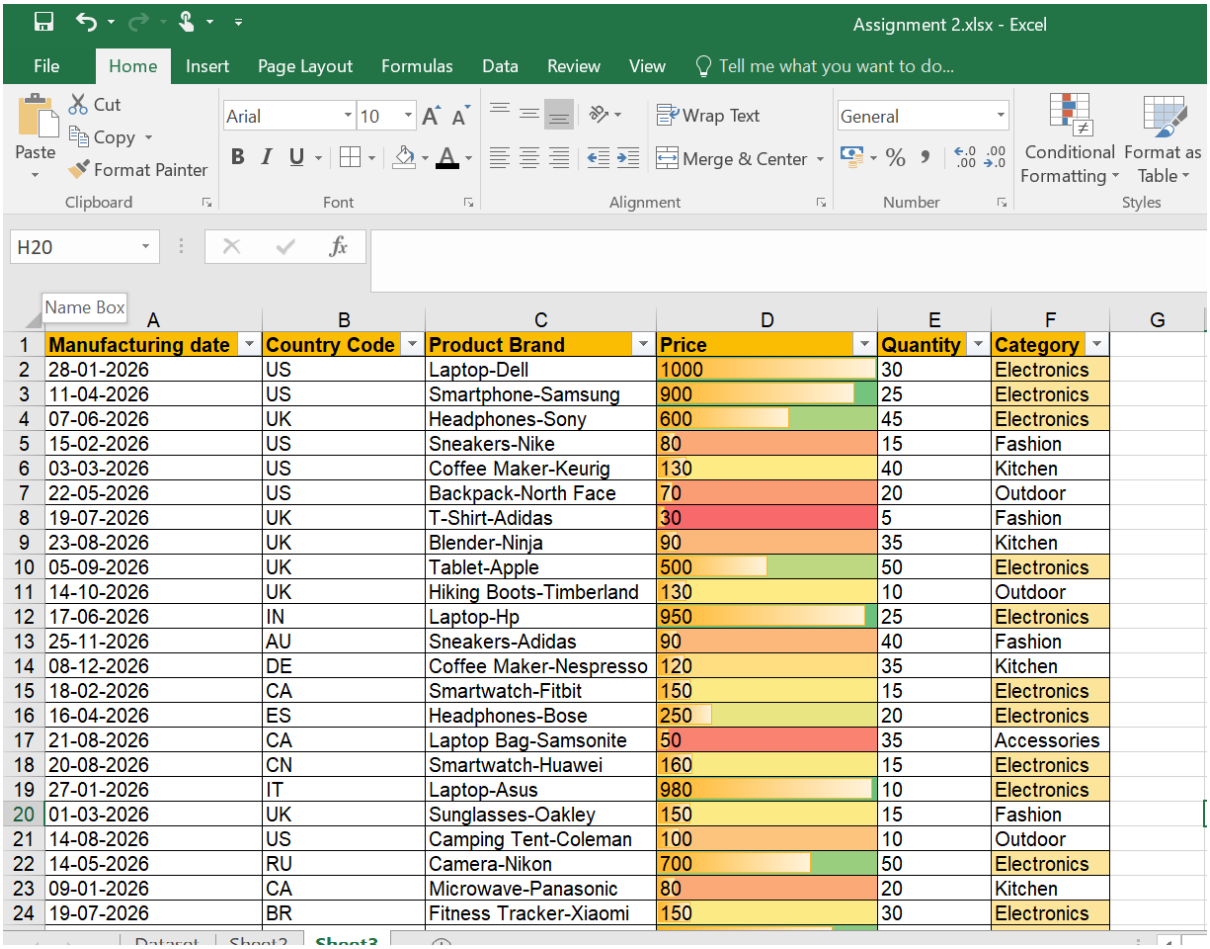
Number Formatting:

- Format the data type of the "Price" column to a currency format.
- Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

Conditional Formatting:

- Apply data bar or color scales conditional formatting in the "Price" column.

- Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



	A	B	C	D	E	F	G
1	Manufacturing date	Country Code	Product Brand	Price	Quantity	Category	
2	28-01-2026	US	Laptop-Dell	1000	30	Electronics	
3	11-04-2026	US	Smartphone-Samsung	900	25	Electronics	
4	07-06-2026	UK	Headphones-Sony	600	45	Electronics	
5	15-02-2026	US	Sneakers-Nike	80	15	Fashion	
6	03-03-2026	US	Coffee Maker-Keurig	130	40	Kitchen	
7	22-05-2026	US	Backpack-North Face	70	20	Outdoor	
8	19-07-2026	UK	T-Shirt-Adidas	30	5	Fashion	
9	23-08-2026	UK	Blender-Ninja	90	35	Kitchen	
10	05-09-2026	UK	Tablet-Apple	500	50	Electronics	
11	14-10-2026	UK	Hiking Boots-Timberland	130	10	Outdoor	
12	17-06-2026	IN	Laptop-Hp	950	25	Electronics	
13	25-11-2026	AU	Sneakers-Adidas	90	40	Fashion	
14	08-12-2026	DE	Coffee Maker-Nespresso	120	35	Kitchen	
15	18-02-2026	CA	Smartwatch-Fitbit	150	15	Electronics	
16	16-04-2026	ES	Headphones-Bose	250	20	Electronics	
17	21-08-2026	CA	Laptop Bag-Samsonite	50	35	Accessories	
18	20-08-2026	CN	Smartwatch-Huawei	160	15	Electronics	
19	27-01-2026	IT	Laptop-Asus	980	10	Electronics	
20	01-03-2026	UK	Sunglasses-Oakley	150	15	Fashion	
21	14-08-2026	US	Camping Tent-Coleman	100	10	Outdoor	
22	14-05-2026	RU	Camera-Nikon	700	50	Electronics	
23	09-01-2026	CA	Microwave-Panasonic	80	20	Kitchen	
24	19-07-2026	BR	Fitness Tracker-Xiaomi	150	30	Electronics	